Counting Particles (template) — teacher key

Question: How do we count things too small to see?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	4	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can say what a mole is.
- I can turn grams into moles with molar mass.
- I can turn moles into a count of particles.

Predictions

- One mole of water and one mole of iron. Which holds more particles? **The same number.** A mole is a count. One mole of anything is 6.02×10^{23} particles.
- Same two samples: one mole of water and one mole of iron. Which has more mass? **Iron.** Same count, but each iron atom is heavier than a water molecule (56 g/mol vs 18 g/mol).

Huddle script (answers in green)

Definitions

- **Mole:** The amount that holds **6.02×10^{23}** particles.
- **Molar mass:** The **mass** of one mole, in grams per mole.

Worked example

How many moles are in 36 g of water? (M = 18 g/mol)

- n = m / M
- n = 36 / 18 = **2** mol
- Particles: $2 \times 6.02 \times 10^{23} =$ **12.04×10^{23}**

Label the water molecule

1. The **oxygen** atom.
2. A **hydrogen** atom.
3. The **bond** that holds them.

Every level, with answers

Grams to moles

Level 1. A sample of water has a mass of 54 g. Water has a molar mass of 18 g/mol. How many moles is that?
Answer: 3 mol n = m / M = 54 / 18 = 3 mol.

Level 2. A sample of carbon has a mass of 36 g. Its molar mass is 12 g/mol. How many moles is that?
Answer: 3 mol n = 36 / 12 = 3 mol.

Level 3. A sample of CH? has a mass of 4 g. The molar mass of CH? is 16 g/mol. How many moles is the sample?
Answer: 0.25 mol n = 4 / 16 = 0.25 mol.

Level 4. You weigh out 116 g of a compound with M = 58 g/mol. How many moles do you have? Answer in mol.
Answer: 2 mol n = 116 / 58 = 2 mol.

Moles to particles

Level 1. How many particles are in 2 mol? Give the answer in units of 10^{23} (so 6.02×10^{23} is written 6.02).
Answer: 12.04×10^{23} $2 \times 6.02 = 12.04$, so 12.04×10^{23} particles.

Level 2. How many particles are in 0.5 mol? Answer in units of 10^{23}.
Answer: 3.01×10^{23} $0.5 \times 6.02 = 3.01 \times 10^{23}$.

Level 3. How many particles are in 4 mol? Answer in units of 10^{23}.
Answer: 24.08×10^{23} $4 \times 6.02 = 24.08 \times 10^{23}$.

Level 4. How many water molecules are in 54 g of water (M = 18 g/mol)? Answer in units of 10^{23}.
Answer: 18.06×10^{23} n = 54 / 18 = 3 mol. Particles = $3 \times 6.02 = 18.06 \times 10^{23}$.

Common misconceptions

Students say	Say instead
A mole is a mass.	A mole is a count: 6.02×10^{23} things.
Multiply the mass by the molar mass.	Divide. Grams ÷ (grams per mole) leaves moles.
More mass always means more particles.	Compare moles, not grams: heavier particles weigh more for the same count.